

ASTRONOMICAL TIMER ASSP

Energy management from dusk to dawn – automatically

Renesas ASSP device TIMR-IS1 is a digital IC integrating the astronomical calculator. The configuration input parameters for the astronomical function are configured via the input pins.

Built-in Astro function takes geographic location inputs from the provided switch inputs and calculates the Sunrise and Sunset timings for the entire Indian & sub-continent with 1204 cities.

Applications

Domestic applications: Gate Lights, Outside lights, Parking, Home Automation

Commercial Applications: Street Illuminations, Road Sign Illuminations, Shopping Center Illuminations (Luminous Signboards), Park / Garden Illuminations, Parking Illuminations

Others: Home Illuminations, Other illumination applications

Technical Specifications

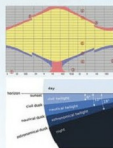
- Supply Voltage (1 Phase): 110 to 240 VAC
- Power Consumption (Max.): 4VA
- Battery Backup : Minimum 5 years running reserve
- LED Indication: Red LED for Relay Status
- Contact Arrangement : 1 C/O (SPDT)
- Contact Rating: 16A (NO) and 5A (NC) @240VAC/ 24VDC
- Incandescent Lamps: 1000 W
- Daily Inductive Load (Cos ϕ = 0.6): 6A @ 250VAC
- Clock Accuracy: Features RTC with built-in 32.768KHz DTCXO, High Stability (UB $\pm 5.0 \times 10^{-6}$ / -40°C to +85°C (Equivalent to 13secs per month deviation)

Features/ Characteristics

- Allows energy saving via Programming of time and geographical area of installation
- Random On Time Delay (up to 59sec) for street light mode
- Daily automatic regulation of sunrise and sunset time.
- Suitable to program the coordinates of the city through easy interface via dip switches
 - Latitude and longitude information is stored for 1204 cities – DIP switch 1
 - City can be selectable through dip switch – DIP switch 2
 - Off time can be programmable through dip switch – DIP switch 3 enabling energy management.
- Separate (RTC) clock battery is present can be programmable through UART
- Manual on/off is present
- Enabling energy management



Off time Setting - Dip S/w 3	
DAWN	--> 0 0 0 0 0 0
10 : 30	--> 0 0 0 0 0 1
10 : 45	--> 0 0 0 0 1 0
11 : 00	--> 0 0 0 0 1 1
11 : 15	--> 0 0 0 0 0 1
11 : 30	--> 0 0 0 1 0 1
11 : 45	--> 0 0 0 0 1 1
0 : 00	--> 0 0 1 1 1 1
0 : 15	--> 0 0 0 0 0 1
0 : 30	--> 0 0 1 0 0 1
0 : 45	--> 0 0 1 0 1 1
1 : 00	--> 0 0 1 1 0 1
1 : 15	--> 0 0 0 0 1 1
1 : 30	--> 0 0 1 0 1 1
1 : 45	--> 0 0 0 1 1 1
2 : 00	--> 0 0 1 1 1 1
2 : 15	--> 0 0 0 0 0 1
2 : 30	--> 1 0 0 0 0 1
2 : 45	--> 0 0 1 0 0 1
2 : 00	--> 1 1 1 0 0 1
3 : 15	--> 0 0 0 1 0 1
3 : 30	--> 1 0 1 0 0 1
3 : 45	--> 0 0 1 1 0 1
4 : 00	--> 1 1 1 0 0 1
4 : 15	--> 0 0 0 0 1 1
4 : 30	--> 1 0 0 0 1 1
4 : 45	--> 0 1 0 0 1 1
5 : 00	--> 1 1 0 0 1 1
5 : 15	--> 0 0 1 1 1 1
5 : 30	--> 1 0 1 1 1 1



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